

Secure AI Systems — TryHackMe Write-Up

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Platform: TryHackMe

Module: Secure AI Systems

Status: Completed

Overview

The Secure AI Systems module provides a defender-focused introduction to securing AI and machine learning systems. It explains how AI components expand the enterprise attack surface, how attackers target these systems, and how to apply structured threat modelling to identify weaknesses. The module emphasizes that AI systems require specialized security considerations beyond traditional application security.

Learning Objectives

By completing this module, I learned how to:

- Understand AI systems as an attack surface
- Identify weaknesses in AI/ML pipelines and integration points
- Recognize LLM-specific security threats
- Apply STRIDE, OWASP, and ATLAS frameworks to AI threat modelling
- Perform reconnaissance on AI systems by identifying exposed ML services and metadata
- Conduct a structured threat modelling assessment on an AI-powered application

Module Sections

1. Securing AI Systems

This section introduces how AI components fit into enterprise architectures and how trust boundaries shift when AI is added. It covers secure design principles, data flow mapping, and the importance of identifying attack surfaces using frameworks such as OWASP and ATLAS.

2. LLM Security

This section focuses on large language models as a unique attack surface. Topics include prompt manipulation, hallucinations, data leakage, and misuse of model capabilities. The module highlights the need for guardrails, monitoring, and human oversight.

3. AI Threat Modelling

This section teaches how to apply STRIDE, OWASP, and ATLAS to AI systems. It reinforces evaluating inputs, outputs, model behavior, data pipelines, and integration points to identify threats and recommend mitigations.

4. AI System Reconnaissance

This section demonstrates how attackers and defenders can discover AI components by scanning for ML services, identifying frameworks, extracting metadata, and reviewing exposed endpoints. It reinforces the importance of minimizing information leakage.

5. AI Threat Modelling Assessment

The module concludes with an interactive assessment where threats are identified, trust boundaries are mapped, and mitigations are recommended. This validates understanding of AI-specific risks and secure design.

Key Takeaways

- AI systems introduce new attack surfaces that traditional controls may not cover
- LLMs require special handling due to prompt manipulation and data leakage risks
- Threat modelling is essential for understanding AI system behavior and integration
- Reconnaissance on AI systems is often easier than expected due to metadata exposure
- Secure AI requires a combination of architecture, controls, monitoring, and governance

Conclusion

- The Secure AI Systems module provides a strong foundation for understanding how AI/ML systems expand the enterprise attack surface. It reinforces the importance of secure design, threat modelling, and continuous monitoring—skills that directly support SOC, security engineering, and governance roles.